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JOSHUA S. SPEAGLE (沈佳士)

Astronomy & Astrophysics | Statistical Sciences
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RESEARCH INTERESTS

My research lies at the intersection of **astrostatistics**, **data science**, and **artificial intelligence**. I develop and apply a variety of methods to **large datasets** to answer complex astrophysical questions and improve how we learn about the world from noisy, messy data.

POSITIONS

Assistant Professor of Astrostatistics: University of Toronto 2022-Present
Research Scientist: River Run Trading Jan-Apr 2025
Quantitative trading strategies for crypto assets
Dunlap Postdoctoral Fellow: University of Toronto 2020-2022
Banting Postdoctoral Fellow: University of Toronto 2020-2022
Supervisor: Gwen Eadie
Project Academic Support Staff: Kavli IPMU, University of Tokyo 2015-2016
Supervisors: Naoki Yoshida, Alexie Leauthaud, & Kevin Bundy

EDUCATION

Harvard University: PhD in Astronomy 2016-2020
Advisers: Doug Finkbeiner, Charlie Conroy, & Daniel Eisenstein (with Alyssa Goodman)
Harvard University: MA in Astronomy 2016-2020
Advisers: Daniel Eisenstein & Alexie Leauthaud
Harvard University: BA in Astrophysics and Physics 2011-2015

SELECTED AWARDS & HONORS

Connaught New Researcher Award: University of Toronto 2024
John Charles Polanyi Prize in Physics: Council of Ontario Universities 2022
Best Astrostatistics Student Paper Award: Astrostatistics Interest Group (ASA) 2020
Eric R. Keto Prize for Best Thesis in Theoretical Astrophysics: Harvard University 2020

SUPERVISION & MENTORSHIP

I am currently (co-)supervising/mentoring a total of **15 individuals**. This includes:
8 postdoctoral researchers
6 doctoral students
1 undergraduate student

See my **website** for a full record of the **85 individuals** I have (co-)supervised/mentored.

PUBLICATIONS

I am an author on **145+ papers** that have **19,000+** citations (h-index=46). This includes:

15 papers as primary/co-first author
27 other papers with significant contributions (non-overlapping categories)
13 papers led by postdocs
22 papers led by students

Most of my papers can be found online on ADS, arxiv, and Google Scholar. My ORCID is 0000-0003-2573-9832. A full list of publications and additional details can be found on my **website**.

SELECTED PROFESSIONAL ACTIVITIES & SERVICE

Professional Societies:

Statistical Society of Canada (SSC): Data Science & Analytics Section President / Past / Elect 2024-2027
American Astronomical Society (AAS): WGAA Steering Committee 2020-2026
LSST Informatics & Statistics Science Collaboration (ISSC): Publication Committee Chair 2024
American Statistical Association (ASA): AIG Program Chair (Interim) 2023-2024
Web Director 2021-2023
Canadian Astronomical Society (CASCA): Postdoc Committee 2020-2023

University Service

Astronomy & Astrophysics: Undergraduate Curriculum Committee (2023-2024, 2025-Present), SURP Committee (Chair 2022-2024, Member 2021-2022, 2026), Training & Mentoring Committee (Co-Chair 2021-2023), Graduate Admissions Committee (2021)
Statistical Sciences: MScAC Data Science (Interim Director 2026-2027), Graduate Admissions Committee (2022-Present), Undergraduate Curriculum Committee (2025-Present), UTSSRP (2026)

Discussion Groups, Workshops, & Conferences

Annual UofT STATSTRO Workshop: Founder & Co-Organizer 2021-Present
Globular Clusters & Their Tidal Tails: SOC Co-Chair, LOC Co-Chair 2024
CASCA Annual General Meeting: SOC Co-Chair, LOC Member 2024

Manuscript Referee

AAS Journals, MNRAS, A&A, RASTI, JOSS, Annals of Applied Statistics, Statistical Science

SELECTED PRESENTATIONS & APPEARANCES

I have given numerous presentations related to work being done by me and my collaborators to the scientific community and the public along with providing lectures and tutorials and participating in panel discussions. See my **website** for a full record.

TEACHING

I have a strong interest in education and pedagogy, with a focus on skills such as programming, applied statistics, and data science. See my **website** for additional details on courses I have taught.

EQUITY, DIVERSITY, & INCLUSION

I am committed to improving equity, diversity, and inclusion (EDI) in the classroom, in my work, and in the wider academic community. See my **website** for additional details.